

A Comparative Analysis of ML Models for Breast Cancer Diagnosis

STSCI 6520 Capstone Project Proposal

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1. Motivation

The early and accurate diagnosis of breast cancer is a critical task in medical treatment. Using Machine Learning (ML) methods can significantly increase accuracy alongside clinical judgement. The Breast Cancer Wisconsin (Diagnostic) Dataset provides a high-dimensional snapshot of tumor characteristics which is applicable to multiple ML methods. Comparing methods in terms of accuracy gives the best potential to diagnose cancer as benign or malignant.

While predictive accuracy is the goal for ML methods it is not the only goal in medical treatment. Doctors and especially patients may not even use the decision of a machine model if they don't understand how it makes a prediction. This is the reason for this proposed study; comparing "black box" models that have high accuracy but exceed intuitive understanding to simpler models that can be interpreted, but may not leverage full data variability for accurate performance.

This project addressed the two-way tradeoff between accuracy and interpretability. In addition to performance measured by accuracy there are additional metrics that can communicate how a model decision is prioritized. This also has implications for how ML terms can be interpreted differently by ML scientists and medical decision makers. For example, **false negative** (a malignant tumor classified as benign) is regarded badly in ML and medicine and the 'dangerous' implication of missing cancer may lead to decisions that include risky surgeries with dangers of organ damage or infection on aging patients. Metrics **Precision** (of all patients the model flagged as "malignant," how many actually were?) and **Recall** (of all true malignant cases, how many did the model find?) may actually be better at giving intuitive understanding for people making medical decisions - but will those terms be understood, especially in relationship to the complexity of the model? This project seeks to find the model that provides the best *balance* between predictive power and interpretability for this critical problem.

2. Research Problem

This project will use the **Breast Cancer Wisconsin (Diagnostic) Dataset** (BCWDD) to cross-compare three classification models. ML classification is frequently used but often not well understood especially by laypersons due to algorithmic complexity and black box methods. Comparing models on the same dataset provides a more intuitive understanding of key features in model implementation and evaluation, like validation and verification, parameter tuning, and comparison metrics. ML methods are becoming more prevalent and more accessible via advanced AI tools; it's worthwhile to learn to compare and question the criteria for a model - even for those who won't build them. Cancer diagnoses and decisions around critical medical

care like surgery or chemotherapy are common to many people. Is language like “low probability of false positives” really interpretable to people caring for loved ones? What other ways of communicating could be considered?

The models under selection are fully detailed in the next section, briefly, they are; **Random Forest**, for accuracy and robustness, **Lasso Regression** selected for interpretability, and **Support Vector Machines**, as a non-linear comparison to Regression. The model performance will be compared on accuracy and interpretability on a meaningful medical dataset.

3. Proposed Steps and Expected Outcomes

All analysis will be conducted using the **R** programming language. The BCWDD data contains the target variable; **diagnosis**, and 30 numerical predictors derived from tumor imaging data, (radius, texture, density etc...). The predictors are numeric and will require preprocessing steps like numeric standardization and mean centering. Best practice ML methods will be used for each - including k-fold cross validation (k=10; 10 groups to fit models). The predictions from each classification model can generate a Confusion Matrix, the number of true and false, misses and matches against the actual diagnoses. These numbers then create the metrics that can be compared for model performance; Accuracy, Precision, and Recall.

The expected outcome of the study is that “black box models” Random Forest and SVM will have the highest performance – but also highly complex and non-linear relationships due to the many numeric predictors. This means the models will lack *interpretability*. Comparatively, Lasso Regression will likely be the most interpretable, but lack the accuracy of the other methods. What remains to be seen is if the potential gap in accuracy is worth providing better understanding for people making decisions.

References

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- [3] Wu, J., et al. (2018). *Logistic LASSO regression for the diagnosis of breast cancer using clinical demographic data and the BI-RADS lexicon for ultrasonography*.